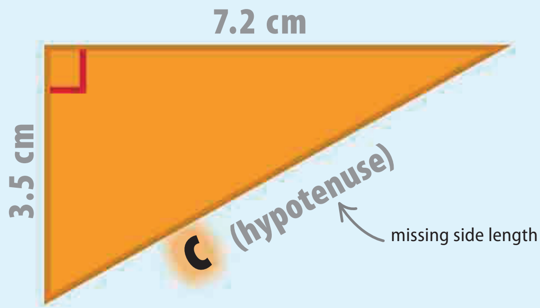
Find the value of the hypotenuse

The Pythagorean Theorem can be used to find the value of the length of the longest side (the hypotenuse) in a right triangle when the lengths of the two shorter sides are known. This example shows how this works, if the two known sides are 3.5 cm and 7.2 cm in length.



$$a^2 + b^2 = c^2$$

First, take the formula for the Pythagorean Theorem.

one side — other side — hypotenuse missing

$$3.5^2 + 7.2^2 = c^2$$

Substitute the values given into the formula, in this example, 3.5 and 7.2.

3.5 × 3.5 equals — 7.2 × 7.2 equals

$$12.25 + 51.84 = c^2$$

Calculate the squares of each of the triangle's known sides by multiplying them.

12.25 + 51.84 equals

$$64.09 = c^2$$

Add these answers together to find the square of the hypotenuse.

sign means square root

$$\sqrt{64.09} = \sqrt{c^2}$$

the square root of 64.09 is the same as the square root of c^2

Use a calculator to find the square root of 64.09. This gives the length of side c.

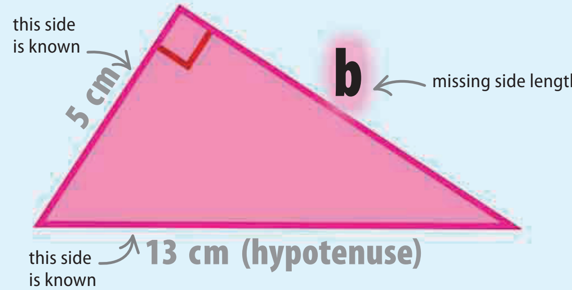
The square root is the length of the hypotenuse.

answer rounded to the hundredths place

$$c = 8.01 \text{ cm}$$

Find the value of another side

The theorem can be rearranged to find the length of either of the two sides of a right triangle that are not the hypotenuse. The length of the hypotenuse and one other side must be known. This example shows how this works with a side of 5 cm and a hypotenuse of 13 cm.



$$a^2 + b^2 = c^2$$

To calculate the length of side b, take the formula for the Pythagorean Theorem.

known side — hypotenuse

$$5^2 + b^2 = 13^2$$

unknown side

Substitute the values given into the formula. In this example, 5 and 13.

unknown side is now result of equation

$$13^2 - 5^2 = b^2$$

hypotenuse now at start of formula

Rearrange the equation by subtracting 5^2 from each side. This isolates b^2 on one side because $5^2 - 5^2$ cancels out.

$$169 - 25 = b^2$$

13 × 13 equals — 5 × 5 equals

Calculate the squares of the two known sides of the triangle.

$$144 = b^2$$

Subtract these squares to find the square of the unknown side.

sign means square root

$$\sqrt{144} = \sqrt{b^2}$$

the square root of 144 is the same as the square root of b^2

Find the square root of 144 for the length of the unknown side.

The square root is the length of side b.

length of missing side

$$b = 12 \text{ cm}$$